```
var = 10
```

```
_var = 10
```

```
num123 = 50.00
```

```
*char = "Mohit"
```

```
  Cell In[81], line 1
    *char = "Mohit"
    ^
SyntaxError: starred assignment target must be in a list or tuple
```

```
1alph = 'Mohit'
```

```
  Cell In[73], line 1
    1alph = 'Mohit'
    ^
SyntaxError: invalid decimal literal
```

```
mohit shastri = "Mohit"
```

```
  Cell In[75], line 1
    mohit shastri = "Mohit"
           ^
SyntaxError: invalid syntax
```

```
mohit123shastri = "Mohit"
```

```
mohit_shastri = "Mohit"
```

Start coding or generate with AI.

```
# Integers ==> int ==> e.g (1,10,33,67,99,100,500)
# Float ==> float ==> e.g (1.0,2.50,10.515,99.90)
# String ==> str ==> e.g ("Mohit","Shastri","RnW","Ahmedabad","10","10.50","True","False","0","1","2+5j","5+10j")
# Complex ==> complex ==> e.g (2+5j -- 2 is Real Part and 5 is Imaginary Part)
# Boolean ==> bool ==> e.g (True / 1, False / 0)
# None ==> Empty/Missing Values
```

```
bool("Mohit")
```

```
True
```

```
bool("")
```

```
False
```

```
bool("-")
```

```
True
```

```
bool(1)
```

```
True
```

```
bool(0)
```

```
False
```

```
bool(-1000)
```

```
True
```

```
bool(000000.1)
```

```
True
```

```
bool(2+5j)
```

```
True
```

```
bool(0+1j)
```

```
True
```

```
bool(0+0j)
```

```
False
```

```
bool(0j)
```

```
False
```

```
bool("0j")
```

```
True
```

```
bool("0")
```

```
True
```

```
bool(None)
```

```
False
```

```
bool("if")
```

```
True
```

```
var = 12
print(var,type(var))
```

```
12 <class 'int'>
```

```
var1 = 12.20
print(var1,type(var1),id(var1))
```

```
12.2 <class 'float'> 2580174224688
```

```
my_str = "Mohit"
print(my_str,type(my_str))
```

```
Mohit <class 'str'>
```

```
a = 2+5j
print(a,type(a))
```

```
(2+5j) <class 'complex'>
```

```
complex(5,-7)
```

```
(5-7j)
```

```
str = "100"
print(str,type(str))
```

```
100 <class 'str'>
```

```
# list ==> [] ==> Datatypes can be any mix
```

```
l1 = ["Mohit",10,20,30.00,2+5j,True,False,1,0,None,10,20,30,"Mohit",True]
print(l1,type(l1))
```

```
['Mohit', 10, 20, 30.0, (2+5j), True, False, 1, 0, None, 10, 20, 30, 'Mohit', True] <class 'list'>
```

```
# tuple ==> () ==> Datatypes can be any mix
```

```
t1 = ("Mohit",10,20,30.00,2+5j,True,False,1,0,None,10,20,30,"Mohit",True)
print(t1,type(t1))
```

```
('Mohit', 10, 20, 30.0, (2+5j), True, False, 1, 0, None, 10, 20, 30, 'Mohit', True) <class 'tuple'>
```

```
# Dictionary ==> dict ==> {} ==> {key:value} pairs
```

```
d1 = {"a":[100,400,500],"b":200,"c":300}
print(d1,type(d1))
```

```
{'a': [100, 400, 500], 'b': 200, 'c': 300} <class 'dict'>
```

```
d1['a']
```

```
[100, 400, 500]
```

```
d1['c']
```

```
300
```

```
d1['b'] = 500
```

```
print(d1,type(d1))
```

```
{'a': [100, 400, 500], 'b': 500, 'c': 300} <class 'dict'>
```

```
print(d1.values())
```

```
dict_values([[100, 400, 500], 500, 300])
```

```
print(d1.keys())
```

```
dict_keys(['a', 'b', 'c'])
```

```
s1 = {"Mohit",10,20,30.00,2+5j,True,False,1,0,None,10,20,30,"Mohit",True}
print(s1,type(s1))
```

```
{False, True, None, 10, (2+5j), 20, 30.0, 'Mohit'} <class 'set'>
```

```
print("Hello" , end = " & ")
print("World")
```

```
Hello & World
```

```
print("Hello")
print("World")
```

```
Hello
World
```

```
print("Hello" , sep = " & ")
print("World")
```

```
Hello
World
```

```
a=10.0
b=10.0
c=a+b
print(c,type(c))
```

```
20.0 <class 'float'>
```

```
a = input()
print(a,type(a))
```

```
 10
10 <class 'str'>
```

```
a = input("Enter a Value : ")
b = input("Enter second Value : ")
c = a+b
print(c,type(c))
```

```
Enter a Value :  10
Enter second Value :  10
1010 <class 'str'>
```

```
a = str(input("Enter a Value : "))
b = str(input("Enter second Value : "))
c = a+b
print(c,type(c))
```

```
Enter a Value :  Mohit
Enter second Value :  Shashtri
MohitShashtri <class 'str'>
```

```
a = int(input("Enter a Value : ")) # int()
b = int(input("Enter second Value : "))
c = a+b
print(c,type(c))
```

```
Enter a Value :  10
Enter second Value :  10
20 <class 'int'>
```

```
a = float(input("Enter a Value : "))
b = float(input("Enter second Value : "))
c = a+b
print(c,type(c),id(c))
```

```
Enter a Value :  10
Enter second Value :  10
20.0 <class 'float'> 1653590150832
```

```
id(a)
```

```
1655194883888
```

```
a = "Hello"
```

```
b = "Hello"
```

```
id(a)
```

```
1653593604672
```

```
id(b)
```

```
1653593604672
```

```
a is not b
```

```
True
```

```
x = "Mohit"
```

```
y = "Mohit"
```

```
id(x)
```

```
2580717816208
```

```
id(y)
```

```
2580717816208
```

```
a = 10.50000001
```

```
b = 10.500001
```

```
id(a)
```

```
1854855981936
```

```
id(b)
```

```
1855415692304
```

Start coding or generate with AI.

```
# int()
# float()
# complex()
# str()
# bool()
# list()
# tuple()
# dict()
# set()
# frozenset()
```

```
a='mohit'
```

```
bool(a)
```

```
True
```

```
complex(2.5,6.5)
```

```
(2.5+6.5j)
```

```
a=5
b=5
```

```
id(a)
```

```
140707726907960
```

```
id(b)
```

```
140707726907960
```

Start coding or generate with AI.

Start coding or generate with AI.

```
a = int(input("Enter a no : "))
b = int(input("Enter a no : "))
a+=1
b+=1
print(a,b)
```

```
Enter a no :  10
Enter a no :  20
11 21
```

```
a = a+1

a+=1
```

```
a = input("Enter a no : ")
b = input("Enter a no : ")
```

```
x = a.isalnum() and b.isalnum()
y = a.isalnum() or b.isalnum()
z = not (a.isalnum() and b.isalnum())
print(x,y,z)
```

```
Enter a no :  zebra
Enter a no :  tiger
True True False
```

```
a = input("Enter a no : ")
b = input("Enter a no : ")
x = a is b and a is not b
y = a is b or a is not b
z = not (a is b and a is not b)
print(x,y,z)
```

```
Enter a no :  zebra
Enter a no :  zebra
False True True
```

```
a = int(input("Enter a no : "))
b = int(input("Enter a no : "))
x = a > b and a == b
y = a < b or a == b
z = not (a == b and a >= b)
print(x,y,z)
```

```
Enter a no :  10
Enter a no :  9
False False True
```

Start coding or generate with AI.